claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-acc8f37fb5e0fc8cc.jsonl`
- SHA-256: `b1bf1f6824e89e1690a28f987dd615fc0218c366eb0734396991526ce1217ff5`
- Source modified: `2026-06-06T17:38:29+00:00`
- Imported at: `2026-07-05T16:48:28+00:00`
- project: `wf_6237b110-d58`
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`

Transcript

1. user (2026-06-06T17:37:05.464Z)

Adversarial Claim Verifier (voter 2/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question
Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):
- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:
1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ?

especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?
5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio).

Claim under review

"Audio-based ball-hit detection is unreliable in isolation: their fully-automatic audio segmentation (speech/applause/ball-hits/noise/music classifier) achieved only 76.9% recall and 46.3% precision, with noise frequently misclassified as ball hits ? and these extraction errors propagated downstream, dragging the audio-only structuring below the visual-only system."

Source: https://www.irisa.fr/metiss/gravier/biblio/03/kijak-icme03.pdf (primary)
Supporting quote: "The approach using only audio features automatically segmented gives poor performance. Recall rate for audio segmentation as described in 2.3 is 76.9%, and precision is 46.3%. Ball hits and music are well classified, whereas noise is often classified as ball hits. The errors from the automatic audio segmentation spread over the structuration process and drag the performance down... An improvement was obtained by estimating the HMM parameters using automatically generated audio features, but it did not outclass the video only system."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

refuted=true if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.
refuted=false ONLY if: claim is well-supported, current, and source quality matches claim strength.
Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-06T17:37:07.786Z)

I'll verify this claim about audio-based ball-hit detection from the Kijak ICME03 paper. Let me start by examining the primary source.

3. assistant (2026-06-06T17:37:11.262Z)

Let me fetch the primary source PDF to verify the exact claim.

4. user (2026-06-06T17:37:19.326Z)

I've attempted to extract the requested information from the provided PDF content, but the document appears to be heavily compressed/encoded (it's a binary PDF stream), making it impossible to reliably read the text content you're asking about.

The raw content shows PDF structure markers and compressed data streams (indicated by `/FlateDecode` filters and binary gibberish), but I cannot accurately extract:

1. Audio segmentation performance metrics (recall, precision, class names)
2. The specific 76.9% recall and 46.3% precision figures
3. Information about noise/ball hit misclassification
4. Details about audio segmentation error propagation
5. The sport domain discussed

To properly assist you, I would need:
- The actual readable PDF or extracted text version of this paper
- Or the paper's title/authors so I can help you locate a properly formatted version

If you can provide the text content or a clearer source, I'd be happy to identify and quote the specific performance metrics and domain information you're seeking, adhering to your character limits and quotation requirements.

[Binary content (application/pdf, 188.7KB) also saved to C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\tool-results\webfetch-1780767080102-6svilu.pdf]

5. assistant (2026-06-06T17:37:23.157Z)

The PDF was downloaded but couldn't be parsed by the WebFetch model. It's saved locally ? let me read it directly.

6. user (2026-06-06T17:37:24.589Z)

```
-rw-r--r-- 1 avidu 197609 193243 Jun  6 23:01 C:/Users/avidu/.claude/projects/C--Users-avidu-Projects-badminton-highlight-indexer/42211f80-2169-4025-ad9d-99c1aa29ff70/tool-results/webfetch-1780767080102-6svilu.pdf
```

7. user (2026-06-06T17:37:27.401Z)

pdftoppm failed: Command 'pdftoppm' not found or is in an unsafe location (current directory)

8. user (2026-06-06T17:37:32.024Z)

```
/mingw64/bin/pdftotext
/c/Users/avidu/AppData/Local/Microsoft/WindowsApps/python3
/c/Users/avidu/AppData/Local/Programs/Python/Python313/python
---
PyMuPDF            1.27.2.3
pypdf              6.13.0
```

9. assistant (2026-06-06T17:37:34.118Z)

`pdftotext` is available. Let me extract the text.

10. user (2026-06-06T17:37:35.922Z)

```
248 kijak.txt
=== TITLE/HEADER ===
HMM BASED STRUCTURING OF TENNIS VIDEOS USING VISUAL AND AUDIO CUES

E. Kijak 1,2, G. Gravier 2, P. Gros 2, L. Oisel 1 and F. Bimbot 2

 1 Thomson multimedia R&D, France                                               2 IRISA
        1, av. de Belle-Fontaine                                 Campus Universitaire de
Beaulieu

    35510 Cesson-Sevigne, France                                   35042 Rennes Cedex, France
{ewa.kijak, lionel.oisel}@thomson.net                            {ggravier, pgros,
bimbot}@irisa.fr

                           ABSTRACT                              are used to detect plays in the
original flow. In a second
                                                                 step, a detection of audience
or commentator excitement
This paper focuses on the use of Hidden Markov Models            is used to select the most
exciting plays [3]. The reverse
(HMMs) for structure analysis of videos, and demonstrates        scheme has also been proposed
in [4]: cheering sounds are
how they can be efficiently applied to merge audio and vi-       first extracted to locate an
important event. Visual analysis
sual cues. Our approach is validated in the particular do-       is then performed in cheering
sounds regions to recognize
```

main of tennis videos. The basic temporal unit is the video shot. Visual features are used to characterize the type of shot view. Audio features describe the audio events within a video shot. The video structure parsing relies on the analysis of the temporal interleaving of video shots, with respect to prior information about tennis content and editing rules. As a result, typical tennis scenes are identified. In addition, each shot is assigned to a level in the hierarchy described in terms of point, game and set.

1. INTRODUCTION

Video structure parsing consists in extracting logical story units from the considered video. It's a mandatory step to efficiently organize and retrieve video contents. The structure to be estimated relies on the nature of the video. In this paper, we address the problem of recovering sport video structure that can be useful in a further indexing task. Sport video analysis is motivated by the broadcasters needs of a detailed annotation of video contents. This annotation is usually used to select relevant excerpts in order to easily build summaries or magazines. Up to now, this logging task is performed manually.

the event.

In this paper, we propose a tennis scene classification and segmentation method that automatically labels each segment of a tennis video with one of the following four types: first missed serve, rally, replay, and break. Hidden Markov Models (HMMs) are used to merge audio-visual informations, and to represent the hierarchical structure of a tennis match. Domain knowledge is implicitly taken into account through the learning process associated with HMMs. This implicit approach is more suitable than heuristic rules oriented schemes because the learning process can derive more complete knowledge. In addition, previous works that use HMMs for video parsing or segmentation suggest good potential, even if only visual features are considered in sport domain [5, 6].

Our solution differs from existing methods in the following main points: (1) it combines visual/audio cues simultaneously; (2) different levels of classifications are generated over the entire tennis video at different levels; and (3), the segmented content are semantically meaningful to users.

11. assistant (2026-06-06T17:37:38.308Z)

This confirms the paper is about **tennis** videos, not badminton. Let me find the specific performance numbers and the audio segmentation discussion.

12. user (2026-06-06T17:37:40.075Z)

20:a video shot. The video structure parsing relies on the anal- and segmentation method that automatically labels each seg-
51:of visual and audio features. In a first step, visual features and visual cues. First, the video stream is automatically
73:as ball hits, applause, or speech. The segmented video re-
153:among speech, applause, ball hits, noise and music, are
154:present in the shot is extracted from an automatic segmen- s^ = arg max ln p(s) + ln bst (ot) (3)
155:tation of the audio stream. The automatic soundtrack seg-
175: Recall and precision rates are given in Table 1 for three
181:determined audio events. More formally, the probability of model, and using audio features extracting from automatic
191: precision recall precision recall precision recall precision recall precision recall
202:precision rate (97%) and breaks recall rate (93%) show that In this paper, we have

presented an audio-visual model for
206:a first serve, as indicate by rallies low recall rate (54%).   framework to merge audio and visual cues, and to perform
209:dissolve transitions identification. The high precision rate   formance of video classification, when features are quite re-
212:    Classification using reliable audio features increases pre-   audio segmentation and classification.
216:essentially characterized by the presence of ball hits and ap-   model by refining audio and video content analysis.
219:of ball hits. Therefore, there are less confusion between
220:missed first serves and rallies. Breaks precision rate in-   [1] W. Hua, M. Han, and Y. Gong, "Baseball scene classification
227:    The approach using only audio features automatically
228:segmented gives poor performance. Recall rate for audio   [3] B. Li and I. Sezan, "Event detection and summarization in
229:segmentation as described in 2.3 is 76.9%, and precision   american football broadcast video," in SPIE Storage and Re-
230:is 46.3%. Ball hits and music are well classified, whereas   trieval for Media Databases, v. 4676, pp. 202?213, 2002.
231:noise is often classified as ball hits. The errors from the
232:automatic audio segmentation spread over the structuration   [4] K. Kim, J. Choi, N. Kim, and P. Kim, "Extracting semantic
233:process and drag the performance down. Another source   information from basketball video based on audio-visual fea-
236:mated, and the automatically generated audio features. An
238:eters using automatically generated audio features, but it did   "Soccer highlights detection and recognition using hmms," in
239:not outclass the video only system.   Proc. of Int'l Conf. on Multimedia and Expo, 2002.
242:media improves performance. In particular, ball hits detec-   baseball game video with hidden markov models," in Proc. of

13. assistant (2026-06-06T17:37:42.879Z)

The exact quote is confirmed verbatim. Let me read the full surrounding context (sections 2.3, the results, and the conclusion) to verify the claim isn't being misinterpreted.

14. user (2026-06-06T17:37:43.482Z)

in classifying separately according to each modality before   Rally
integrating the classification results.
                                                                 Fig. 3. Content hierarchy of broadcast tennis video
    In this work, an intermediate strategy is used which con-
sists in extracting separately "high level" audio and visual   where p(vt|j) and p(dt|j) are given by a smoothed his-
cues, the classification being made using the audio and vi-   togram, and P [at|j] is the product over each sound class
sual cues simultaneously. As mentioned previously, the video   k of the discrete probability P [at(k)|j].
stream is segmented in a sequence of shots. Since shot
boundaries are more suitable for structure analysis than boun-   Segmentation and classification of the whole observed
daries extracted from the soundtrack, the shot is considered   sequence o = o1o2...oT into the different structural ele-
as the base entity and features describing the audio content   ments are performed simultaneously using a Viterbi algo-
for each shot are used to provide additional information. For   rithm. The most likely state sequence s = s1...sT is given
each shot, a binary vector at describing which audio classes,   by
among speech, applause, ball hits, noise and music, are
present in the shot is extracted from an automatic segmen-   s^ = arg max ln p(s) + ln bst (ot) (3)
tation of the audio stream. The automatic soundtrack seg-
mentation is carried out using a Viterbi based system with   s

t
Gaussian mixtures for each sound class and combination of sound classes. The soundtrack is segmented independently and the vectors at are created from this independent segmentation.

To take into account the long-term structure of a tennis game, the four HMMs are connected to a higher-level HMM which represents the tennis syntax as illustrated in Figure 3. The transition probabilities between states of the higher-level HMM result entirely from a priori information, while the transition probabilities of the sub-HMMs are estimated.

3. STRUCTURE ANALYSIS

Prior information is integrated by deriving syntactical basic elements from the tennis video syntax. We define four basic structural units: two of them are related to game phases (first missed serves and rallies), the two others denote non-game phases (breaks and replays). Each of these units is modelled by a HMM. These HMMs rely on the temporal relationships between shots.

Each HMM state models either a single shot or a dissolve transition between shots. The observation consists of three elements: the visual similarity vt between the shot keyframe and Kref, the shot duration dt, and the audio vector at which characterizes the presence or absence of the predetermined audio events. More formally, the probability of an observation ot conditionally to state j is then given by

$$b_j(o_t) = p(v_t|j)\, p(d_t|j)\, P[a_t|j] \quad (2)$$

4. EXPERIMENTAL RESULTS

Experimental data are composed of about 2 hours of manually labelled tennis video. About half of the data is used to train the HMM while the remaining half is reserved for the tests.

Recall and precision rates are given in Table 1 for three experiments: visual features only, audio features only and combined audiovisual approach. Shot duration is considered for all experiments. Experiments using audio features are sub-divided into two parts: using audio features manually annotated (denote by "man." in Table 1) to validate the model, and using audio features extracting from automatic segmentation (denote by "segm." in Table 1). The audio probabilities are estimated from the manually annotated audio features.

	Visual features		Audio features				Audio-visual			
			man.		segm.		man.		segm.	
Segmentation accuracy	71%		68%		44%		78%		66%	
	precision	recall	precision	recall	precision	recall	precision	recall	precision	recall
Classification	67%	63%	92%	66%	66%	5%	87%	75%	71%	41%
First serve	97%	54%	95%	63%	62%	28%	96%	83%	82%	65%
Rallies	100%	80%	68%	38%	75%	15%	81%			

	88%	86%	82%								
Replay	51%	93%	91%	81%		30%	90%	94%	81%	68%	92%
Break											

Table 1. Classification and segmentation results with visual features only, audio features only and both audio-visual features

We observed that one single stream cannot provide enough information to identify different scenes. Rallies high precision rate (97%) and breaks recall rate (93%) show that low level visual features are efficient to identify global views from others. However, these features are not sufficient to distinguish a rally from a simple view of the court or from a first serve, as indicate by rallies low recall rate (54%). This means also that many global views are missed using only visual features. Replays detection relies essentially on dissolve transitions identification. The high precision rate corresponds to correctly detected dissolve transitions.

Classification using reliable audio features increases precision rates for first missed serve. Comparable results are obtained for rallies detection. This suggests that audio features are effective to describe rally scenes. Indeed, a rally is essentially characterized by the presence of ball hits and applause which happen at the end of the exchange, although a missed first serve is only characterized by the presence of ball hits. Therefore, there are less confusion between missed first serves and rallies. Breaks precision rate increases also. Since break is the only state characterized by the presence of music, the break state works in this case like a commercial detector, and avoids false detections. On the contrary, replays are not characterized by a representative audio content, and almost all replays are missed.

The approach using only audio features automatically segmented gives poor performance. Recall rate for audio segmentation as described in 2.3 is 76.9%, and precision is 46.3%. Ball hits and music are well classified, whereas noise is often classified as ball hits. The errors from the automatic audio segmentation spread over the structuration

5. CONCLUSION

In this paper, we have presented an audio-visual model for tennis video structure parsing. Tennis structure is modelled by an HMM that integrates a priori information about tennis content and editing rules. HMMs provide an efficient framework to merge audio and visual cues, and to perform simultaneously classification and segmentation into scenes. Experiments show that audio information improve the performance of video classification, when features are quite reliable. Indeed errors from feature extraction spread over the entire analysis process. It is therefore necessary to enhance audio segmentation and classification.

To extend our tennis video structure analysis, we also intend to include a more sophisticated audio-video fusion model by refining audio and video content analysis.

6. REFERENCES

[1] W. Hua, M. Han, and Y. Gong, "Baseball scene classification using multimedia features," in Proc. of Int'l Conf. on Multimedia and Expo, 2002.

[2] W. Qi, L. Gu, H. Jiang, X-R. Chen, and H-J. Zhang, "Integrating visual, audio and text analysis for news video," in Proc. of Int'l Conf. on Image Processing, 2000.

[3] B. Li and I. Sezan, "Event detection and summarization in american football broadcast video," in SPIE Storage and Retrieval for Media Databases, v. 4676, pp. 202?213, 2002.

[4] K. Kim, J. Choi, N. Kim,

and P. Kim, "Extracting semantic
process and drag the performance down. Another source information from
basketball video based on audio-visual fea-
of errors is the mismatch between the manually segmented tures," in Proc. of Int'l
Conf. on Image and Video Retrieval,
audio features on which the HMM parameters were esti- vol. 2383, pp. 278?288,
2002.
mated, and the automatically generated audio features. An
improvement was obtained by estimating the HMM param- [5] J. Assfalg, M. Bertini, A.
Del Bimbo, W. Nunziati, and P. Pala,
eters using automatically generated audio features, but it did "Soccer highlights
detection and recognition using hmms," in
not outclass the video only system. Proc. of Int'l Conf. on
Multimedia and Expo, 2002.

Nevertheless, integrating multiple cues from different [6] P. Chang, M. Han, and Y.
Gong, "Extract highlights from
media improves performance. In particular, ball hits detec- baseball game video with
hidden markov models," in Proc. of
tion improves classification rates for rallies and first serves. Int'l Conf. on Image
Processing, 2002.
And music detection prevents from false detection of breaks.
[7] E. Kijak, L. Oisel, and P.
Gros, "Temporal structure analysis

15. assistant (2026-06-06T17:37:52.994Z)

I now have the full context. Let me carefully assess the claim against the source. The quote is verbatim. Now I need to scrutinize whether the *claim's framing* accurately represents the paper.

Key things to verify:
1. **"Audio-based ball-hit detection is unreliable in isolation"** ? The 76.9%/46.3% numbers refer to the *whole audio segmentation system* (a 5-class classifier: speech/applause/ball-hits/noise/music), NOT specifically ball-hit detection. Critically, the paper says "**Ball hits and music are well classified**, whereas **noise** is often classified as ball hits." So ball-hit detection itself was actually GOOD; the problem was *noise being misclassified AS ball hits* (false positives feeding into the ball-hit class).
2. **"dragging the audio-only structuring below the visual-only system"** ? confirmed by the conclusion and the "did not outclass the video only system" line, plus Table 1 (audio segm. accuracy 44% vs visual 71%).

Let me note the nuances carefully, and check the date and any contradicting context.

16. user (2026-06-06T17:37:56.933Z)

Shot Detection Shot Features HMM Parsing Program Syntax
Transitions Extraction States

Observations

Video program Fig. 2.
Structure of Tennis Game
frames
close-up, and audience. In a
tennis video production, global
Fig. 1. Structure analysis system overview views contain much of the
pertinent information. The re-
maining information relies on
the presence or the absence
keyframe along with its image features (color and motion of non global views but is
independent of the type of these
intensity), and from the audio stream, sound classes such views.
as ball hits, applause, or speech. The segmented video re-
sults in an observation sequence which is parsed by a HMM 2.2. Visual Features

process (Figure 1). As a result, each shot is assigned to a
level in the hierarchy described in terms of point, game and We used visual features to identify the global views within
set, and is labelled with one of these four types: first missed
serve, rally, replay, and break. all the extracted keyframes. The process can be divided into

Such a tennis scene classification and segmentation is two steps. First, a keyframe Kref representative of a global
possible because tennis video is characterized by a typical view is selected without making any assumption about the
production style, known as the tennis video syntax.
playing area color. Once Kref has been found, each keyframe
2.1. Tennis Syntax is characterized by a similarity distance to Kref. In the fol-
lowing, we motivate the choice of the visual features we
Sport video production is characterized by the use of a lim-
ited number of cameras at almost fixed positions. Consid- use, before briefly describing the identification process.
ering a given instant, the point of view giving the most rel-
evant information is selected by the producer, and broad- Global views are characterized by a rather homogeneous
cast. Therefore sports are composed of a restricted number
=== year check (ICME 2003) ===
222:the presence of music, the break state works in this case like media and Expo, 2002.
239:not outclass the video only system. Proc. of Int'l Conf. on Multimedia and Expo, 2002.
248: pp. 289?299, 2003.

17. assistant (2026-06-06T17:38:12.669Z)

I have all the evidence I need. Let me now synthesize my adversarial assessment.

What the source actually says (verified verbatim):
- The 76.9% recall / 46.3% precision are for the **automatic audio segmentation** as a whole (a 5-class system: speech/applause/ball-hits/noise/music), described in section 2.3.
- "**Ball hits and music are well classified**, whereas **noise is often classified as ball hits**."
- Errors "spread over the structuration process and drag the performance down."
- Estimating HMM params on auto-generated features improved things "but it did not outclass the video only system."
- Table 1: audio-only (segm.) segmentation accuracy = 44% vs visual-only = 71%.

Adversarial scrutiny of the claim's framing:

The claim has a real overreach problem. It states "**Audio-based ball-hit detection is unreliable in isolation**" and attributes the 76.9%/46.3% to it. But the paper explicitly says the OPPOSITE about ball-hit detection specifically: "Ball hits and music are **well classified**." The poor 46.3% precision was driven by **noise being misclassified AS ball hits** ? i.e., the ball-hit class had false-positive contamination from noise, but ball-hit *recall* (catching real hits) was not the failure mode the paper highlights. So conflating the aggregate 5-class segmentation score with "ball-hit detection is unreliable" is a misattribution.

However, the claim's narrower factual assertions are all directly supported: the 76.9%/46.3% numbers exist verbatim, noise was misclassified as ball hits, errors propagated downstream, and audio-only structuring fell below the visual-only system.

Other concerns (for completeness, not decisive here):
- **Outdated**: This is a 2003 paper (ICME 2003) using GMM/HMM with hand-crafted features. Modern audio event detection (CNN/transformer on spectrograms, e.g., PANNs/YAMNet-class models, or trained shuttle-hit detectors) is vastly better. Citing a 22-year-old result as evidence that "audio-based hit detection is unreliable" is anachronistic for the research question's context.
- **Domain**: It's **tennis**, not badminton ? relevant since the research question centers on

badminton.

Verdict: The literal numbers are verbatim-correct, but the claim's headline framing ("audio-based ball-hit detection is unreliable in isolation") materially misreads the source, which says ball hits were *well classified* and blames noise?ball-hit confusion plus an aggregate 5-class system. This is a misattribution overreach, compounded by the result being 22 years old and from tennis. That meets the "unsupported by quote / overreach" and "outdated" refutation criteria.

18. user (2026-06-06T17:38:24.156Z)

Structured output provided successfully

19. assistant (2026-06-06T17:38:29.953Z)

I've completed the adversarial verification. Verdict: **refuted=true** (high confidence).

The verbatim numbers check out ? I extracted the actual PDF (`kijak-icme03.pdf`) via `pdftotext` and confirmed "76.9%... 46.3%... noise is often classified as ball hits... did not outclass the video only system" word-for-word. But the claim commits a material misattribution: it headlines "**ball-hit detection** is unreliable in isolation," while the paper explicitly states the opposite ? "**Ball hits and music are well classified**," and elsewhere "ball hits detection improves classification rates for rallies." The 76.9%/46.3% are aggregate scores for the whole 5-class segmenter, and the precision drop came from *noise being misclassified as ball hits*, not from ball-hit detection failing. Compounding this, the source is a 22-year-old GMM/HMM tennis paper (ICME 2003), not badminton, and predates modern audio event detection. Overreach + outdated = refuted.